Military Specification
Transmitting Set, Radio AN/PRT-5 (*)

This amendment forms a part of Military Specification MIL-R-23449A(AS), 28 July 1969, and has been approved by the Naval Air Systems Command, Department of the Navy.

Page 2, Paragraph 2.1, under Naval Air Systems Command: Add

"AR-34	Failure Classification for Reliability Testing, General Requirements for"

Page 10, Paragraph 4.4.1.3: Delete and Substitute:

4.4.1.3 Manufacturing Run in Test- Each equipment shall be operated under the conditions specified herein for a period of ten hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least ten hours of operation

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of ten minutes prior to the beginning of the ten hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the ten hour period of operation following the ten minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be

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repaired and the test started over, except that the ten minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The ten hour period specified above may be composed of two five hour periods to conform with standard working hours.

Page 11, Paragraph 4.4.1.3.1: Delete

Page 11, Paragraph 4.4.1.3.2: Delete

Page 12, Paragraphs 4.4.2 thru 4.4.2.4: Delete and Substitute:

4.4.2 Reliability Assurance Tests- Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34.

4.4.2.1 Qualification Phase- Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level II shall be used. The Accept-Reject Criteria for Test Plan E shall be used, except that the lower test temperature shall be -30°C.

4.4.2.2 Reliability Production Acceptance (Sampling) Phase Tests- The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests." Test level E shall be used except that the lower temperature limit shall be -30°C.

4.4.2.2.1 All Equipment Test- Each equipment produced except those submitted for the Reliability Qualification Test, shall be tested for 50 hours. Prior to the 50 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included in the approved test procedures. The 50 hours of operation shall be failure free. Any equipment experiencing failure during the 50 hour test shall be repaired and returned to test for another 50 hour period of operation so that the last 50 hours of operation are failure free. To determine whether the MTBF is being met at any time during the contract the total operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the curves of Test Plan II of MIL-STD-781. (Extend the lines as necessary to accomodate the data.) At no time shall the plot of total test time and failures cross the accept

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line (extended) of Test Plan II. These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.2.3 Length of Heat Portion of Cycle- After stabilization at the high temperature limit required by test level E (+55C) the equipment shall be operated 3.0 hours during each cycle (as per 5.2.3.2 of MIL-STD-781B).

4.4.2.4 Performance Characteristics to be Measured- The following equipment characteristics are to be measured during the stabilized heat portion of the cycle:

(1) Modulation (per 3.4.4)
(2) Frequency Stability (per 3.4.2)
(3) Power Output (per 3.4.5)

4.4.2.5 Test Details- The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests."

Preparing Activity:
NAVY - AS

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